

Permutations of an array

arr = {1, 2, 3} n = 3

1 2 3

1 3 2

2 1 3

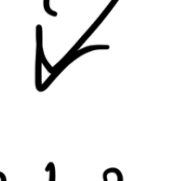
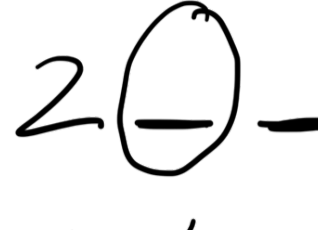
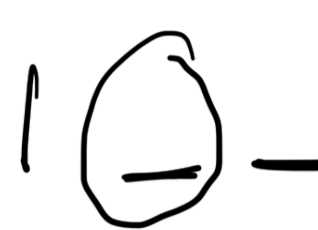
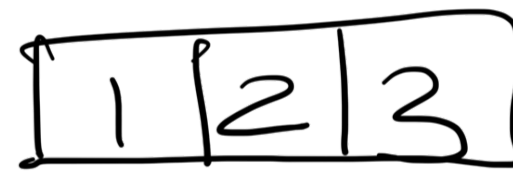
2 3 1

3 1 2

3 2 1

$$6 = n! = 3!$$

n choices n-1 choices n-2 choices ... = n!



1 2 3

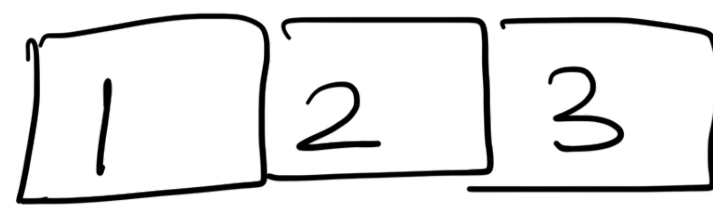
1 3 2

2 1 3

2 3 1

3 1 2

3 2 1



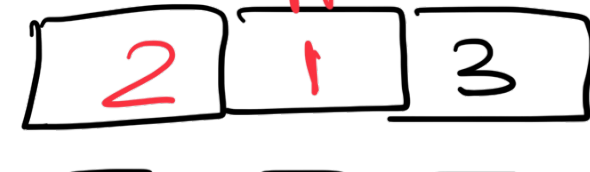
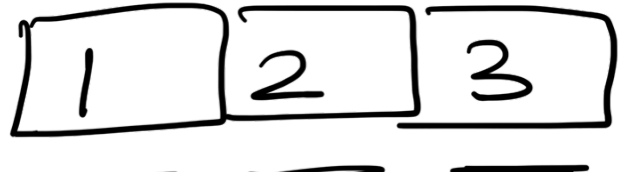
0 1 2

idn

no swap

swap

swapped



idn = 1

idn = 1

idn = 1

no swap

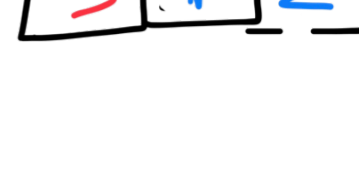
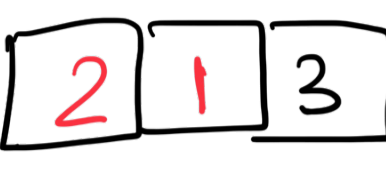
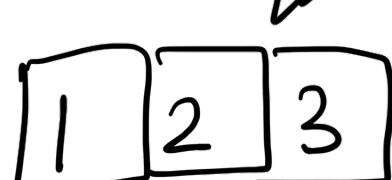
swap

no

yes

no

yes



Code

```
if(idn == n) ans.pb(nums) } base
return } case
```

```
for(i = idn; i < n; i++) {
```

```
    swap(nums[idn], nums[i])
```

```
    getPerms(nums, idn + 1, ans)
```

```
    swap(nums[idn], nums[i])
```

```
}
```

//backtracking step we need to make sure that the array that we swapped before going down reverts back to original before going up so we can take the right choice ie choose 2 or 3 at 0th position wala step

TC: n! calls x n choices

$$TC = O(n! \times n)$$

SC: n! + call stack (n levels deep)

$$O(n! + n) \Rightarrow O(n!)$$